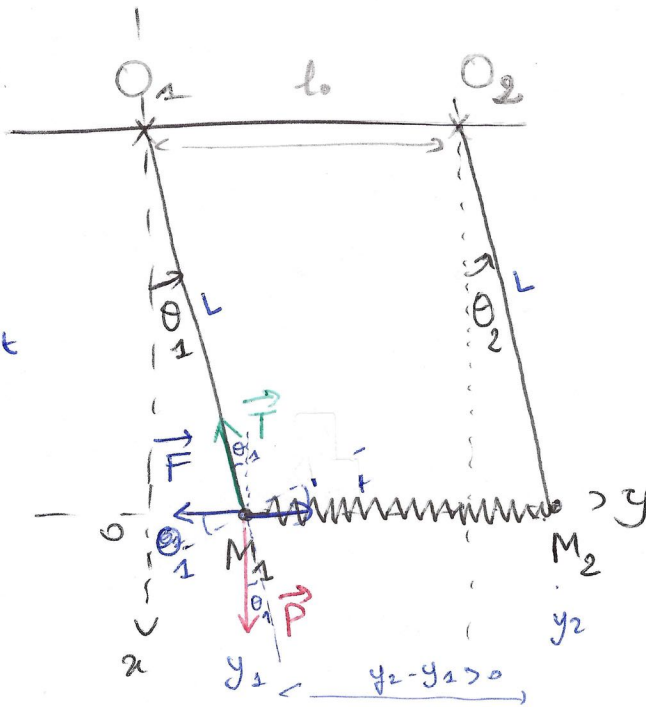


Pendules couplés.



$$y_1 = L\theta_1$$

$$y_2 = L\theta_2 + l_0$$

$$\vec{F} = -k \Delta l$$

$$\vec{F} = -k \vec{uy} (l_0 - l)$$

$$\vec{F} = -k \vec{uy} (l_0 - (y_2 - y_1))$$

$$= -k (l_0 - y_2 + y_1) \vec{uy}$$

$$= k (l_0 - l_0 - L\theta_2 + L\theta_1) \vec{uy}$$

$$\vec{F}_{M_1} = -k L (\theta_1 - \theta_2) \vec{uy}$$

③ TMC

$$\frac{dL_{O_1}}{dt} = \sum_i \vec{m}_i (\vec{r}_i \times \vec{F}_i)$$

2 m

$$\vec{OM_1} \wedge m \vec{v_1} = L \times m L \dot{\theta}_1 \vec{u_z} = mL^2 \dot{\theta}_1 \vec{u_z}$$

②

$\vec{M}(T) = \vec{0}$ car coupe l'axe

3

$$\vec{M}_{O_1}(\vec{P}) = \vec{O_1 M_1} \wedge \vec{P} = -L \times mg \times \sin \theta_1 \vec{u_z} \quad (\text{fait tourner dans le sens retrograde})$$

$$\vec{M}_{O_1}(\vec{F}) = L \cdot k (l_0 - y_2 + y_1) \times \sin(\frac{\pi}{2} - \theta_1) \vec{u_z}$$

$$= L k (l_0 - y_2 + y_1) \cos \theta_1 \vec{u_z}$$

④ TMC :

$$mL^2 \ddot{\theta}_1 = -mgL \sin \theta_1 - kL (l_0 - y_2 + y_1) \cos \theta_1$$

3

$$mL^2 \ddot{\theta}_1 = -mgL \theta_1 - kL (\theta_1 - \theta_2) \quad \downarrow \theta_1 \ll 1 \text{ rad}$$

$$\ddot{\theta}_1 = -\frac{g}{L} \theta_1 - \frac{k}{m} (\theta_1 - \theta_2)$$

$$\ddot{\theta}_1 + \left(\frac{g}{L} + \frac{k}{m} \right) \theta_1 = \frac{k}{m} \theta_2$$

$$\ddot{\theta}_1 + \omega_0^2 \theta_1 = \Omega_0^2 \theta_2$$

eq' orde 2.

$$⑤ \quad mL\ddot{\theta}_2 = -mgL\sin\theta_2 + kL(\theta_2 - \theta_1)$$

$\downarrow \theta_2 \ll 1 \text{ rad}$

$$mL\ddot{\theta}_2 = -mgL\theta_2 + kL(\theta_2 - \theta_1)$$

$$3 \quad \vec{F}_{m_2} = k\vec{uy}(l-l_0) \quad (= -\vec{F}_{m_1})$$

Les 2 eq sont symétriques.

$$(1) \quad \ddot{\theta}_1 = -\frac{g}{L}\theta_1 - \frac{k}{m}(\theta_1 - \theta_2)$$

$$(2) \quad \ddot{\theta}_2 = -\frac{g}{L}\theta_2 - \frac{k}{m}(\theta_2 - \theta_1)$$

$$\begin{cases} \ddot{\theta}_1 + \omega_0^2\theta_1 = \Omega_0^2\theta_2 & (1) \\ \ddot{\theta}_2 + \omega_0^2\theta_2 = -\Omega_0^2\theta_1 & (2) \end{cases}$$

2. 6) On pose $S(t) = \begin{pmatrix} \theta_1(t) \\ \theta_2(t) \end{pmatrix}$ et $D(t) = \theta_1(t) - \theta_2(t)$

$$(1)+(2) \quad \ddot{\theta}_1 + \ddot{\theta}_2 + \omega_0^2(\theta_1 + \theta_2) = \Omega_0^2(\theta_1 + \theta_2)$$

$$\ddot{S}(t) + \omega_0^2 S = \Omega_0^2 S$$

⑦ et ⑧
2 2

$$\ddot{S} + (\omega_0^2 - \Omega_0^2)S = 0$$

$$\Rightarrow S(t) = S_m \cos(\omega_1 t + \varphi)$$

$$\omega_1 = \sqrt{\omega_0^2 - \Omega_0^2} \text{ défini.}$$

$$(1)-(2) \quad \ddot{\theta}_1 - \ddot{\theta}_2 + \omega_0^2(\theta_1 - \theta_2) = \Omega_0^2(\theta_2 - \theta_1)$$

$$\ddot{D} + \omega_0^2 D = -\Omega_0^2 D$$

$$\ddot{D} + (\omega_0^2 + \Omega_0^2)D = 0$$

$$\Rightarrow D(t) = D_m \cos(\omega_2 t + \varphi)$$

$$\omega_2(t) = \sqrt{\omega_0^2 + \Omega_0^2}$$

$$⑤ \quad S(t) = \Theta_1 + \Theta_2$$

$$2 \quad D(t) = \Theta_1 - \Theta_2$$

$$\Theta_1 = \frac{S+D}{2} \quad \Theta_2 = \frac{S-D}{2}$$

$$\Rightarrow \Theta_1(t) = \frac{1}{2} \left(S_m \cos(\omega_1 t + \varphi_1) + D_m \cos(\omega_2 t + \varphi_2) \right)$$

$$\Theta_2(t) = \frac{1}{2} \left(S_m \cos(\omega_1 t + \varphi_1) - D_m \cos(\omega_2 t + \varphi_2) \right)$$

$$\Theta_1(t) = A \cos(\omega_1 t + \varphi_1) + B \cos(\omega_2 t + \varphi_2)$$

$$\Theta_2(t) = A \dots \dots \dots - B \dots \dots \dots$$

Auch möglich:

$$S(t) = C \cos \omega_1 t + E \sin \omega_1 t$$

$$D(t) = F \cos \omega_2 t + G \sin \omega_2 t$$

$$\Rightarrow \Theta_1(t) = \frac{C}{2} \cos \omega_1 t + \frac{E}{2} \sin \omega_1 t + \frac{F}{2} \cos \omega_2 t + \frac{G}{2} \sin \omega_2 t$$

$$\Theta_2(t) = \frac{C}{2} \cos \omega_1 t + \frac{E}{2} \sin \omega_1 t - \frac{F}{2} \cos \omega_2 t - \frac{G}{2} \sin \omega_2 t$$